

Aim: Implementation of Events, Commands, and Tasks in TinyOS
Develop a TinyOS program that toggles LEDs using events and tasks to demonstrate split-phase execution.

Source Code

1. Module Implementation ([BlinkC.nc](#))

Implements the core application logic, timer events, commands, and deferred task execution.

Code:

```
#include "Timer.h"
module BlinkC @safe() {
  uses interface Timer<TMilli> as Timer0;
  uses interface Leds;
  uses interface Boot;
}
implementation {
  uint16_t counter = 0;

  // TASK: Deferred background execution (Split-Phase Phase 2)
  task void toggleLedTask() {
    counter++;
    dbg("BlinkC", "TASK RUNNING: Processing split-phase task, counter = %d\n", counter);

    // COMMAND: Call LED driver to toggle LED0
    call Leds.led0Toggle();

    // Toggle LED1 on every alternate execution
    if (counter % 2 == 0) {
      call Leds.led1Toggle();
      dbg("BlinkC", "TASK: LED1 toggled on even count.\n");
    }
  }

  // EVENT: System booted
  event void Boot.booted() {
    dbg("BlinkC", "BOOT EVENT: System booted, starting periodic timer.\n");
    // COMMAND: Start 250ms periodic timer
    call Timer0.startPeriodic(250);
  }

  // EVENT: Hardware timer tick (Split-Phase Phase 1)
  event void Timer0.fired() {
    dbg("BlinkC", "EVENT: Timer0 fired! Posting task to TinyOS scheduler...\n");
    // Post task to scheduler queue to avoid blocking interrupt context
    post toggleLedTask();
  }
}
```

2. Configuration & Wiring ([BlinkAppC.nc](#))

Wires the application module interfaces to system components ([MainC](#), [LedsC](#), and [TimerMilliC](#)).

Code:

```
configuration BlinkAppC {  
}  
implementation {  
  components MainC;  
  components LedsC;  
  components BlinkC as App;  
  components new TimerMilliC() as Timer0;  
  
  // Wiring interfaces  
  App.Boot -> MainC.Boot;  
  App.Leds -> LedsC;  
  App.Timer0 -> Timer0;  
}
```

3. Simulation Script ([sim.py](#))

Configures the TOSSIM simulation engine, initializes node 1, routes debug channels to standard output, and executes simulation events.

```
import sys  
from TOSSIM import *  
  
t = Tossim([])  
r = t.mac()  
  
# Route debug channels  
t.addChannel("BlinkC", sys.stdout)  
t.addChannel("LedsC", sys.stdout)  
  
# Boot mote 1 at time 0  
m = t.getNode(1)  
m.bootAtTime(0)  
  
print("=== Starting Split-Phase Execution Simulation ===")  
  
# Run 100 simulation events  
for i in range(100):  
    t.runNextEvent()  
  
print("=== Simulation Finished ===")
```

OUTPUT:

```

[INFO] *** Successfully built micaz TOSSIM library.
aquib@ubuntu:~/tinys-main/apps/Blink$ python sim.py
=== Starting Split-Phase Execution Simulation ===
0:0:0.000000000 DEBUG (1): BOOT EVENT: system booted, starting periodic timer.
0:0:0.244140635 DEBUG (1): EVENT: Timer0 fired! Posting task to TinyOS scheduler...
0:0:0.244140645 DEBUG (1): TASK RUNNING: Processing split-phase task, counter = 1
0:0:0.244140645 DEBUG (1): LEDS: Led0 off.
0:0:0.488281260 DEBUG (1): EVENT: Timer0 fired! Posting task to TinyOS scheduler...
0:0:0.488281270 DEBUG (1): TASK RUNNING: Processing split-phase task, counter = 2
0:0:0.488281270 DEBUG (1): LEDS: Led0 on.
0:0:0.488281270 DEBUG (1): LEDS: Led1 off.
0:0:0.488281270 DEBUG (1): TASK: LED1 toggled on even count.
0:0:0.732421885 DEBUG (1): EVENT: Timer0 fired! Posting task to TinyOS scheduler...
0:0:0.732421895 DEBUG (1): TASK RUNNING: Processing split-phase task, counter = 3
0:0:0.732421895 DEBUG (1): LEDS: Led0 off.
0:0:0.976562510 DEBUG (1): EVENT: Timer0 fired! Posting task to TinyOS scheduler...
0:0:0.976562520 DEBUG (1): TASK RUNNING: Processing split-phase task, counter = 4
0:0:0.976562520 DEBUG (1): LEDS: Led0 on.
0:0:0.976562520 DEBUG (1): LEDS: Led1 on.
0:0:0.976562520 DEBUG (1): TASK: LED1 toggled on even count.
0:0:1.220703135 DEBUG (1): EVENT: Timer0 fired! Posting task to TinyOS scheduler...
0:0:1.220703145 DEBUG (1): TASK RUNNING: Processing split-phase task, counter = 5
0:0:1.220703145 DEBUG (1): LEDS: Led0 off.
0:0:1.464843760 DEBUG (1): EVENT: Timer0 fired! Posting task to TinyOS scheduler...
0:0:1.464843770 DEBUG (1): TASK RUNNING: Processing split-phase task, counter = 6
0:0:1.464843770 DEBUG (1): LEDS: Led0 on.
0:0:1.464843770 DEBUG (1): LEDS: Led1 off.
0:0:1.464843770 DEBUG (1): TASK: LED1 toggled on even count.
0:0:1.708984385 DEBUG (1): EVENT: Timer0 fired! Posting task to TinyOS scheduler...
0:0:1.708984395 DEBUG (1): TASK RUNNING: Processing split-phase task, counter = 7
0:0:1.708984395 DEBUG (1): LEDS: Led0 off.
0:0:1.953125010 DEBUG (1): EVENT: Timer0 fired! Posting task to TinyOS scheduler...
0:0:1.953125020 DEBUG (1): TASK RUNNING: Processing split-phase task, counter = 8
0:0:1.953125020 DEBUG (1): LEDS: Led0 on.
0:0:1.953125020 DEBUG (1): LEDS: Led1 on.
0:0:1.953125020 DEBUG (1): TASK: LED1 toggled on even count.
0:0:2.197265635 DEBUG (1): EVENT: Timer0 fired! Posting task to TinyOS scheduler...
0:0:2.197265645 DEBUG (1): TASK RUNNING: Processing split-phase task, counter = 9
0:0:2.197265645 DEBUG (1): LEDS: Led0 off.
0:0:2.441406260 DEBUG (1): EVENT: Timer0 fired! Posting task to TinyOS scheduler...
0:0:2.441406270 DEBUG (1): TASK RUNNING: Processing split-phase task, counter = 10
0:0:2.441406270 DEBUG (1): LEDS: Led0 on.
0:0:2.441406270 DEBUG (1): LEDS: Led1 off.
0:0:2.441406270 DEBUG (1): TASK: LED1 toggled on even count.
=== Simulation Finished ===
aquib@ubuntu:~/tinys-main/apps/Blink$

```